

CDF-Intervals Revisited

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Abstract. The concept of *cdf*-intervals introduced in previous work is revisited with a new domain specification, inference mechanism and implementation. A *cdf*-interval extends traditional convex intervals over reals with a degree of knowledge attached to the data. Instead of approximating the unknown data distribution with the nearest uniform distribution, leading to a linear *cdf* curve, we bound the unknown probability distribution thus guaranteeing a full encapsulation of the data and its degree of knowledge with two curves. A *cdf*-interval is now specified by two triplets denoting the lower and upper bounds slopes of the data distribution. A triplet is composed of the data value, degree of knowledge and respectively lowest and steepest slopes. This idea is embodied as a new solver in the ECLⁱPS^e system, presented in this paper. Experimental results indicate that the solver brings solution insights, leading to tighter bounds when compared with the previous *cdf* and the convex data interval solvers.

1 Introduction

Interval coefficients have been introduced in Operations Research and Constraint Programming to specify uncertain data in order to provide reliable solutions to convex models. They are at the heart of paradigms such as robust optimization [3, 14] in Operations Research as well as mixed CSP [8], reliable constraint reasoning [23, 24], and quantified CSP [25] in Constraint Programming. These paradigms specify erroneous and incomplete data using uncertainty sets that denote a deterministic and bounded formulation of an ill-defined data. To remain computationally tractable, the uncertainty sets are approximated by convex structures such as intervals (extreme values within the uncertainty set) and interval reasoning can be applied ensuring effective computations. The concept of convex modeling was coined to formalize the idea of enclosing uncertainty sets and yield reliable solutions; i.e guaranteed to contain any solution produced by any possible realization of the data [6, 2, 24]. As a result, the outcome of such systems is a solution set that can be refined when more knowledge is acquired about the data, and does not exclude any potential solution. The benefits of these approaches are that they deal with real data measurements, produce robust/reliable solutions, and do so in a computationally tractable manner. However, the solution set can sometimes be too large to be meaningful since it encloses all

solutions that can be constructed using the data intervals. Furthermore each solution derived has equal uncertainty weight, thus does not reflect any possible degree of knowledge about the data. For instance, consider a collected set of data measurements in traffic flow analysis [11], management of inventory with stochastic demand [21], the data is generally ill-defined but some data values can occur more than others (hence greater degree of knowledge or certainty). This quantitative information is available during data collection, but lost during the reasoning because not accounted for in the representation of the uncertain data. As a consequence it is not available in the solution set produced. Mainly reliable models offer reliability and robustness, tractable models, but do not account for quantitative information about the data.

In [19] a novel extension of the interval data model with a second dimension was introduced: a quantitative dimension added to the measured input data. The main idea was to show that one can preserve the tractability of convex modeling while enriching the uncertain data sets with a representation of the degree of knowledge available. The cumulative distribution function (*cdf*) over the density function was used to represent the degree of knowledge since i) it is monotone, suitable for interval computations and pruning, ii) it *directly* represents the aggregated probability that a quantity lies within bounds, iii) it brings flexibility to the problem modeling assumptions. The limitation of this model is that it does not "bound" the potential actual data distribution curve itself and thus restricts the information gain when reasoning about the uncertainty.

In this paper we revisit this domain representation with a new specification of *cdf*-intervals, exploiting the concept of probability box. Instead of *approximating* the unknown data distribution with the nearest uniform distribution, we bound the complete unknown probability distribution thus guaranteeing a full encapsulation of the data and its degree of knowledge with two linear curves. A *cdf*-interval is now specified by two triplets denoting the lower and upper bounds linear slopes of the data distribution. A triplet is composed of the data value, degree of knowledge and respectively lowest and steepest slopes. This idea is embodied as a new solver in the ECLⁱPS^e system, presented in this paper. Experimental results indicate that the solver brings solution insights, leading to tighter bounds when compared with the previous *cdf* and the convex data interval solvers. This paper contains the following contributions and structure: (1) a new representation and specification of *cdf*-intervals, (2) a framework for solving systems of arithmetic constraints over *cdf*-intervals, (3) an experimental evaluation to network design problems. The experimental work shows how this new domain representation yields tighter domain bounds, and more informed results while remaining computationally effective and competitive with previous works.

2 Basic Concepts

This section recalls fundamental concepts we use to characterize the basic features of our framework. These definitions can be found in [20, 22, 12, 4, 10].

Notations Throughout this paper we assume independence of data, and assume that data takes its value in the set of positive real numbers $\mathbb{R}$, denoted by a, b, c . Data points are denoted by p, q, r possibly subscripted by a data value. Variables are denoted by X, Y, Z and take their values in $\mathcal{U}$. Intervals of elements in $\mathcal{U}$ are symbolized I, J, K .

Cumulative Distribution Function The *cdf* defines its accumulated probability density so far. More formally, $F_X(x)$ is the probability of a variable X to be at most x ; it expresses the distribution of data in an aggregated manner.

Definition 1 (cumulative distribution function). *Given an item value x , with density function $f(x)$, and an unknown variable (commonly referred to as the real-valued random variable) X , the cdf of x $F_X(x)$ is the function:*

$$F_X(x) = \int_{-\infty}^x f_X(x) dx \quad \text{continuous random variable} \quad (1)$$

The cdf ranges over the interval $[0, 1]$. Every cdf is monotone, and right continuous.

The *cdf* of a uniformly distributed random variable over a given interval $[a, b]$ is $F_X(x) = \int_{-\infty}^x \frac{1}{b-a} dx$. Thus, a uniformly distributed *cdf* is graphically represented by a line with slope $\frac{1}{b-a}$ between the two bounds of the interval $[a, b]$. The *cdf* of the variable is 0 for all values below a and saturates at 1 for all quantiles greater than b .

Joint Cumulative Distribution Function Operations can be performed on *cdfs* but carry a different interpretation than operations over standard arithmetic calculus since they relate to probabilities. The joint operation is essential to our solver and is recalled below. The *joint cdf* results from superimposing densities of two variables in a relation, each exerting a *cdf* on its own.

Definition 2 (joint cdf). *Given two random variables X and Y*

$$F_{XY}(x, y) = Pr(X \leq x, Y \leq y) \quad (2)$$

For independent variables $Pr(X \leq x, Y \leq y) = F_X(x) \times F_Y(y)$.

Addition of two randomly distributed variables is operated by convolution. Generally, convolution is an expensive joint pointwise product of densities involved in the addition.

Definition 3 (convolution operation). *Given $z = x + y$, convolution yields the probability density function $f_{XY}(z)$; where both random variables X and Y engaged in the addition take values up to x and y respectively; to obtain the resulting distribution, the intersections³ of densities are accumulated for each value of $z \in (-\infty, \infty)$.*

³ Intersection of probabilities is the product of their densities

$$f_{XY}(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z-x)dx = \int_{-\infty}^{+\infty} f_X(z-y)f_Y(y)dy$$

$$F_{XY}(z) = \int_{X+Y \leq z} f_{XY}(z)dz$$

Stochastic dominance We need to make use of probability distribution orderings. In probability theory the stochastic dominance defines a list of partial ordering on random variables.

Definition 4 (First order stochastic dominance). *Given X and Y , Y dominates X iff*

$$Pr(X > x) \leq Pr(Y > x) \quad \forall x \in (-\infty, +\infty) \quad (3)$$

In terms of *cdfs* we have: $F_Y(x) \leq F_X(x) \quad \forall x \in (-\infty, +\infty)$. In general, *cdfs* curves might not be comparable under the first order stochastic dominance for some quantile x : $F_X(x) \leq F_Y(x)$. We recall the second order stochastic dominance that handles such cases, and is used in our framework.

Definition 5 (Second order stochastic dominance). *A random variable Y has a second order stochastic dominance over X iff:*

$$\int_{-\infty}^x F_Y(x)dx \leq \int_{-\infty}^x F_X(x)dx \quad \forall x \in (-\infty, +\infty) \quad (4)$$

The integration of *cdf* calculates the area under the curve from $-\infty$ to $+\infty$. Second stochastic ordering indicates that the mean of X is at least as high as that of Y .

Probability Box: p-box We now present the notion of p-boxes that will constitute the heart of our new *cdf*-interval domain specification. p-boxes are interval-probabilistic bounds that enclose an imprecisely known probability distribution [9, 22]. A p-box is basically a convex structure over a set of probabilities.

Definition 6 (p-box). *Let $\underline{F}$ and $\overline{F}$, be respectively the dominated and dominant distribution bounds. $[\underline{F}_X, \overline{F}_X]$ specifies the probability box of a random variable X whose distribution F_X is contained within the p-box bounds:*

$$\underline{F}_X(x) \leq F_X(x) \leq \overline{F}_X(x) \quad \forall x \in (-\infty, +\infty) \quad (5)$$

Similar to numeric interval arithmetic, interval-probabilistic arithmetic is applied on random variables. Given two random variables X and Y , $F_X \in [\underline{F}_X, \overline{F}_X]$ and $F_Y \in [\underline{F}_Y, \overline{F}_Y]$ are the uncertain *cdf* of X and Y respectively. The joint distribution of X and Y $\sigma([F_X, F_Y])$ is contained within the joint p-box structure defined by the joint bounds $\sigma([\underline{F}_X, \underline{F}_Y])$ and $\sigma([\overline{F}_X, \overline{F}_Y])$.

$$\sigma([\underline{F}_X, \underline{F}_Y]) \leq \sigma([F_X, F_Y]) \leq \sigma([\overline{F}_X, \overline{F}_Y]) \quad (6)$$

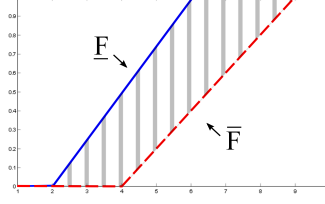


Fig. 1. p-box structure

3 *cdf*-intervals as p-box

In this section we describe how we define and specify our new *cdf*-intervals that denote both uncertain data and variables domains. While the bounds that specify uncertain data are constructed once for all as a preprocessing step, variable domain bounds will be subject to pruning by means of inference rules.

Intuition The core idea behind the new domains is that the whole unknown value and its degree of knowledge are bounded by two linear slopes and associated value using a p-box. We illustrate in Fig. 2 an observed collected data from a measuring instrument, its *cdf* that shapes a staircase function. We display both the linear approximation defined in previous work and the one we propose in this paper as two linear *cdf* bounds that enclose the whole data and its degree of knowledge. One can see that the novel approach does not aim at approximating the curve but rather enclosing it in a reliable manner.

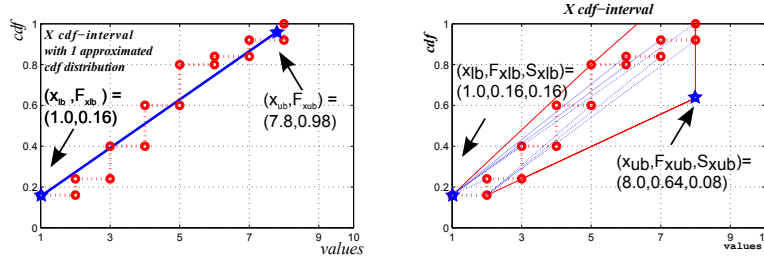
Fig. 2. *cdf*-interval: from a linear approximation to a p-box for X

Fig. 2 illustrates the same variable X that ranges in the *cdf*-interval $[(1, 0.16), (7.8, 0.98)]$ and in the p-box $[(1, 0.16, 0.16), (8.0, 0.64, 0.08)]$. This indicates that quantile value of X is in $[1.0, 8.0]$. The *cdf*-interval representation approximates the distribution that ranges between $[0.16, 0.98]$; on the other hand the p-box bounds the measured distribution by the two bound triplets. Quantile 1.0 has a chance

of occurrence for X that cannot exceed 16%, whilst X can be at most 8.0 with a probability that is at least 64%.

The p-box bounds are defined by two points in a 2D-space. Each bound is a triplet in a 2D-space. Each point p_x is defined by $(x, F_x^p, S_x^p) \in \mathcal{U}$, such that x is the quantile value, F_x^p its *cdf* value, and S_x^p the slope of the *cdf* distribution the point lies on.

Definition 7 (Ordering over $\mathcal{U}, \preceq$). Let $p_x = (x, F_x^p, S_x^p)$, $q_y = (y, F_y^q, S_y^q) \in \mathcal{U}$, the ordering $\preceq$ is a partial order defined by:

$$p_x \preceq q_y \Leftrightarrow x \leq y \text{ and } \int_{-\infty}^y F^q dy \leq \int_{-\infty}^y F^p dy$$

p_x and q_y are located on two different *cdf*-curves: F^p and F^q respectively. The integral ordering signifies that F^q is dominating F^p using the second order stochastic dominance. F^p and F^q are calculated from the stored *cdf* value and slope.

Example 1. Consider three points $p_x = (5.0, 0.22, 0.047)$, $q_y = (7.0, 0.68, 0.09)$ and $r_z = (2, 0.43, 0.68)$ which are depicted in Fig. 3(a). We have $r_z \preceq p_x$ and $r_z \preceq q_y$, but p_x and q_y are not comparable.

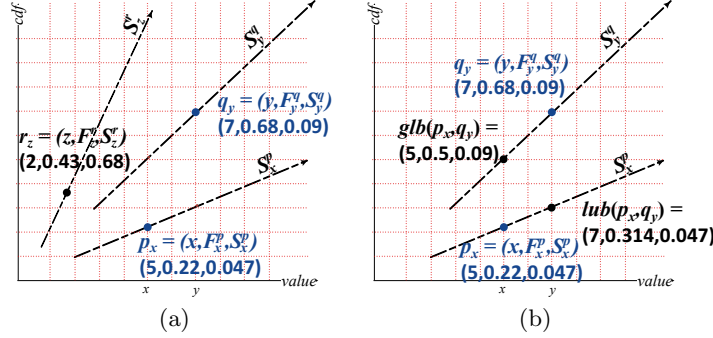


Fig. 3. Ordering over $\mathcal{U}$: (a) points ordering (b) *glb* and *lub* computation

Definition 8 (cdf-interval as p-box). A *cdf-interval* is defined by two points p_x and $q_y \in \mathcal{U}$, denoted $[p_x, q_y]$ such that $p_x \preceq q_y$.

4 p-box domains and propagators

An essential task in interval reasoning is the computation of a new interval from arbitrary points or previous intervals, such that it describes the smallest interval containing a collection of elements. This is based on the meet and joint operators.

Definition 9 (meet and join). *Given the arithmetic ordering and meet and join operations over reals $(\mathbb{R}, \leq, \min, \max)$ and stochastic orderings of cdf distributions, the meet (glb) and join (lub) operators of two points p_x and q_y in $\mathcal{U}$ are defined by:*

$$\begin{aligned} glb(p_x, q_y) &= (\min(x, y), \max(\int_{-\infty}^{lb} F^p dlb, \int_{-\infty}^{lb} F^q dlb)) \\ lub(p_x, q_y) &= (\max(x, y), \min(\int_{-\infty}^{ub} F^p dub, \int_{-\infty}^{ub} F^q dub)) \end{aligned} \quad (7)$$

where $lb = \min(x, y)$ and $ub = \max(x, y)$

The following property establishes the link between the partial ordering $\preceq$ and the pair (glb, lub) as actual meet and join.

Property 1 (Consistency property).

$$\begin{aligned} p_x \preceq q_y &\Leftrightarrow p_x = glb(p_x, q_y) \\ p_x \preceq q_y &\Leftrightarrow q_y = lub(p_x, q_y) \end{aligned} \quad (8)$$

Fig. 3(b) illustrates an example which computes glb and lub of two points $p_x = (5, 0.22, 0.047)$ and $q_y = (7, 0.68, 0.09)$. $glb(p_x, q_y) = (5, 0.5, 0.09)$ and $lub(p_x, q_y) = (7, 0.314, 0.047)$.

A *cdf*-interval is now a p-box that represents a family of probability distributions. Together with real (min, max) operations, the second order stochastic dominance is deployed in our framework to characterize the bounds of the interval. Designated bounds are chosen to follow uniform distributions in order to maintain tractable computations while extending the expressive power of the framework.

Definition 10 (p-box domain). *A p-box domain is a pair $[p_a, p_b]$ satisfying $p_a \preceq p_b$ and denoting the set:*

$$\{p_x = (x, F_x^p, S_x^p) \mid p_a \preceq p_x \preceq p_b\}$$

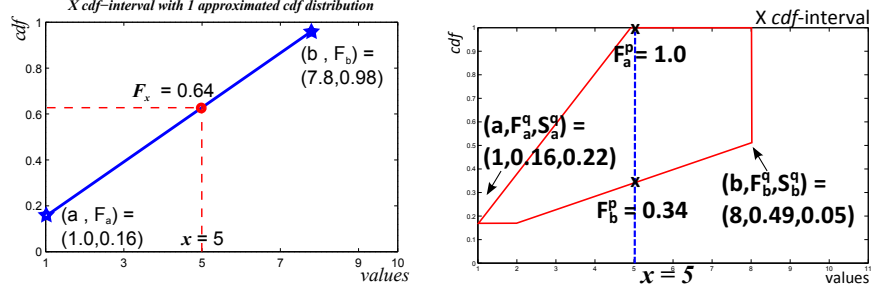
where $a \leq x \leq b$ and $F_a^{p'} \geq F_x^p \geq F_b^{p'}$ and $S_a^p \geq S_x^p \geq S_b^p$

$F_a^{p'}$ and $F_b^{p'}$ are the possible minimum and maximum *cdf* values p_x can take; both are computed by projecting the point p_x onto the *cdf* distributions passing through points a and b respectively and they are derived using the following linear projections:

$$F_a^{p'} = \min(F_a^p - S_a^p(a - x), 1) \quad \text{and} \quad F_b^{p'} = \max(F_b^p - S_b^p(b - x), 0)$$

The smallest convex interval which represents two arbitrary points p_x and q_y defined in $\mathcal{U}$ is $[glb(p_x, q_y), lub(p_x, q_y)]$

A *cdf* variable X is sought to range over a *cdf*-interval domain $I = [p_a, p_b]$ if and only if $X \in [p_a, p_b]$

Fig. 4. computing *cdf*-bounds for $p_x \in I$

Example 2. Consider a *cdf*-variable $X \in I = [(1, 0.16, 0.22), (8, 0.49, 0.05)]$, as illustrated in Fig. 4, X can take any point value (x, F_x^p, S_x^p) meaning that $1 \leq x \leq 8$, $\min(0.16 - 0.22(1 - x), 1) \geq F_x^p \geq \max(0.49 - 0.05(8 - x), 0)$ and $0.22 \geq S_x^p \geq 0.05$. Given a data value $x = 5$ its computed *cdf* lies within the values $[0.34, 1.0]$; this is oppose to the extracted *cdf* approximated value 0.64 which lies within the computed range. Obtaining a *cdf* range enables us to deduce that the maximum possible chance for x to be at most 5 is 100% and its minimum probability of occurrence to be less than 5 is 34%. Closer difference between deduced values designates higher degree of data whereabouts.

4.1 p-box *cdf*-domain calculus

Just like with interval reasoning, we aim at maintaining the convex structure of p-box *cdf*-intervals. Algebraic convex *cdf* structures are defined in the set of ranges $\mathcal{R}_{\mathcal{U}}$ and are ordered by the $\text{cdf} \subseteq_{\mathcal{R}_{\mathcal{U}}}$ operation

Definition 11. Let $I = [p_a, p_b]$ and $J = [q_c, q_d]$ be two *cdf*-interval ranges $\in \mathcal{R}_{\mathcal{U}}$, $\text{cdf} \subseteq_{\mathcal{R}_{\mathcal{U}}}$ is defined by:

$$[p_a, p_b] \subseteq_{\mathcal{R}_{\mathcal{U}}} [q_c, q_d] \Leftrightarrow p_a \preccurlyeq q_c \text{ and } p_b \preccurlyeq q_d$$

Definition 12. Given the ordering, meet and join operations over *cdf*-intervals $(\mathcal{U}, \preccurlyeq, \text{glb}, \text{lub})$, the meet of the interval ranges $\text{glb}_{\mathcal{R}_{\mathcal{U}}}$ yields the intersection of domains under consideration; and the join $\text{lub}_{\mathcal{R}_{\mathcal{U}}}$ defines their union. For two ranges $[p_a, p_b]$ and $[q_c, q_d]$ in $\mathcal{R}_{\mathcal{U}}$ they are defined by:

$$\begin{aligned} \text{glb}_{\mathcal{R}_{\mathcal{U}}}([p_a, p_b], [q_c, q_d]) &= [\text{lub}(p_a, q_c), \text{glb}(p_b, q_d)] \\ \text{lub}_{\mathcal{R}_{\mathcal{U}}}([p_a, p_b], [q_c, q_d]) &= [\text{glb}(p_a, q_c), \text{lub}(p_b, q_d)] \end{aligned} \quad (9)$$

Clearly, the meet operation yields a new domain range with the tightest possible area. This fact guides the decision maker towards obtaining tighter surroundings of the unknown distribution. Conversely the latter operation encloses all possible points of its constituent domain interval ranges.

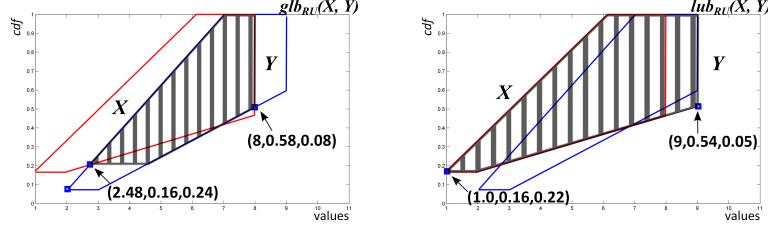


Fig. 5. computing $glb_{\mathcal{R}_U}$ and $lub_{\mathcal{R}_U}$

Example 3. Fig. 5 shows the computation of $glb_{\mathcal{R}_U}$ and $lub_{\mathcal{R}_U}$ over two *cdf* intervals. X and Y are the output of two different measuring instruments interpreting the same set of observations. For $X \in [(1, 0.16, 0.22), (8, 0.49, 0.05)]$ and $Y \in [(2, 0.07, 0.24), (9, 0.67, 0.08)]$ we have:

$$\begin{aligned} glb_{\mathcal{R}_U}(X, Y) &= [(2.48, 0.16, 0.24), (8, 0.58, 0.08)] \\ lub_{\mathcal{R}_U}(X, Y) &= [(1.0, 0.16, 0.22), (9, 0.54, 0.05)] \end{aligned}$$

Arithmetic operations on the interval ranges extend operations interpreted over the set of real intervals and combine uncertainties in the framework. Based on [22] probabilistic results of all possible pairwise relations between two random variables with ill-defined probabilities are contained within the resulting computation exerted on their upper and lower *cdf* distribution bounds.

Definition 13. Consider two intervals $I = [p_a, p_b]$ and $J = [q_c, q_d]$, their arithmetic addition is a result of adding every two points p_x and q_y from both intervals. The resulting interval is specified by

$$[(lb_+, F_{lb_+}^{I+J}, S_{lb_+}^{I+J}), (ub_+, F_{ub_+}^{I+J}, S_{ub_+}^{I+J})]$$

Real interval arithmetic addition is applied to compute lower and upper quantile bounds respectively denoted lb_+ and ub_+

$$lb_+ = \min(a + c, a + d, b + c, c + d) \text{ and } ub_+ = \max(a + c, a + d, b + c, c + d)$$

The resulting *cdf* bounds, $F_{lb_+}^{I+J}$ and $F_{ub_+}^{I+J}$ are obtained by convoluting lower bound distributions (F_a^p and F_c^q) and upper bound distributions (F_b^p and F_d^q) respectively.

$$\begin{aligned} F_{lb_+}^{I+J} &= \frac{1}{2} \frac{(a_{lb} + c_{lb} - lb_+)^2}{(b_{lb} - a_{lb})(d_{lb} - c_{lb})}, S_{lb_+}^{I+J} = \frac{F_{lb_+}^{I+J}}{lb_+ - (a_{lb} + c_{lb})} \\ F_{ub_+}^{I+J} &= 1 - \frac{1}{2} \frac{(b_{ub} + d_{ub} - ub_+)^2}{(b_{ub} - a_{ub})(d_{ub} - c_{ub})}, S_{ub_+}^{I+J} = \frac{1 - F_{ub_+}^{I+J}}{ub_+ - (b_{ub} + d_{ub})} \end{aligned} \quad (10)$$

Empirical evaluation To evaluate the added value of the new constraint domain, we considered an example provided in [24], and attached *cdf* value and slope to the interval bounds. Example 4 aims at solving a system of linear equations which has p-box *cdf*-intervals coefficients and unknown variables. Shown below are pruned domains of the variables at fixed point using inference rules that are established as a separate module in the ECLⁱPS^e constraint programming environment [7]. The p-box *cdf*-intervals attached to the data (here coefficients) were propagated onto the *cdf*-variables, X_1 and X_2 . Inferences on the quantile component yield similar pruning on the resulting variable domains and the additional information coming from the *cdf* and slope components demonstrate the information gained on the density of occurrence for the resulting points within the p-box *cdf*-domains.

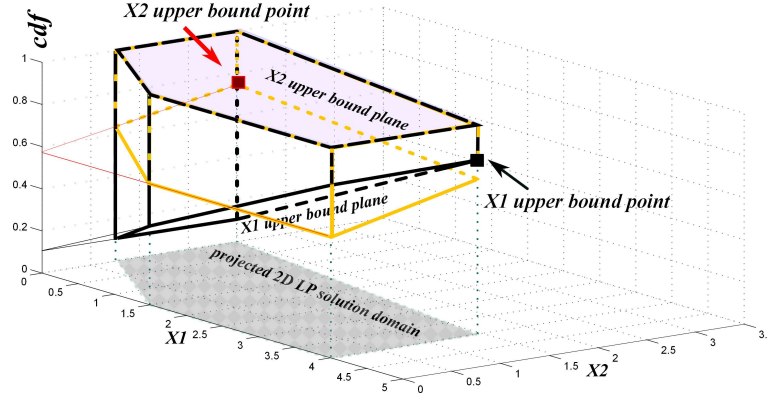


Fig. 6. Example 4: Solution set resulting from the p-box *cdf*-interval computations

Example 4. Consider the system of linear equations $(\mathbf{A}, \mathfrak{R}, \mathbf{b})$ shown below:

$$\mathbf{A} = \begin{bmatrix} [(1.0, 0.95, 0.41), (2.0, 0.41, 0.095)] & [(0.0, 0.83, 0.32), (2.0, 0.32, 0.083)] \\ [(0.0, 0.31, 0.75), (1.0, 0.75, 0.031)] & [(0.0, 0.35, 0.87), (1.0, 0.87, 0.035)] \\ [(1.5, 0.14, 0.96), (3.0, 0.96, 0.034)] & [(5.999, 0.9, 0.86), (6.0, 0.6, 0.028)] \end{bmatrix},$$

$$\mathfrak{R} = \begin{bmatrix} \subseteq_{\mathcal{R}_U} \\ = \\ = \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} [(3.0, 0.88, 0.4), (4.0, 0.4, 0.088)] \\ [(0.0, 0.3, 0.78), (5.0, 0.78, 0.03)] \\ [(4.0, 0.29, 0.85), (15.0, 0.85, 0.029)] \end{bmatrix}$$

In a typical uncertain CSP algebraic structure all values in the resulting solution set have an equally weighted quantitative information. The output is an interval which can be too large and lack the meaning on the data whereabouts. Clearly p-box *cdf*-intervals propagate in the solution set a quantitative information that is coupled with values resulting from the adopted *cdf*-operations. Fig. 6

illustrates two boxes, each encloses the output solution population residence for $X1$ and $X2$. The shown boxes are the result of applying p-box *cdf*-intervals propagation techniques on the linear equations provided in this example. Note that for the original *cdf*-intervals with an approximated uniform distribution solution to the problem yield one plane per variable that is enclosed by the shown p-boxes and which specifies an approximated cumulative uniform distribution of the data whereabouts. The black box is the representation of $X1$ solution p-box domain $[(0.0, 1.0, +\infty), (4.0, 0.667, 0.1669)]$ and the yellow box is the solution domain of $X2$ $[(0.0, 1.0, +\infty), (2.5, 0.598, 0.4344)]$. Clearly, solutions intersect in the $X1 - X2$ 2D-space as illustrated by their projection depicted by the shaded checkerboard region: the real-interval arithmetic solution. This indicates that $X1$ and $X2$ quantiles range respectively between $[0.0, 4.0]$ and $[0.0, 2.5]$. The p-box representation shows a minimum possible chance of occurrence for quantiles to be at most 4.0 and 2.5 is 66.7% and 59.8% respectively for both variables $X1$ and $X2$. Note that quantiles of $X1$ and $X2$ cannot lie outside the p-box bounds defined by the two dimensions: quantiles and cdf and represented by the depicted variable 2D-plane.

4.2 When all constraints are linear

When all the model constraints are linear one can hope to benefit from techniques used in Interval Linear Systems (ILS) [13]. An ILS is a linear model whereby the coefficients are real intervals. When the coefficients are all positive we talk about a POLI system [1]. Such systems can be solved effectively by transforming the POLI into an equivalent linear model with twice the number of constraints. Such systems do not aim at deriving one value for each variable but instead a tighter reliable interval that holds whatever the data realisation might be. The solution set is computed by running two Simplex over each variable (e.g. $\min(X)$ and $\max(X)$ as objective functions) to derive their new bounds. We show that this effective technique can be generalized to p-box *cdf*-intervals.

The algorithm is sketched out below and is composed of five main steps:

1. Generate linear inequalities
2. Extract quantile ILS
3. Solve 2n eplex instances
4. Project the *cdf*-distributions
5. Extract the final solution p-box *cdf*-interval bounds

Definition 14. Let $\mathcal{V}$ be a set of n *cdf*-variables, and $\mathcal{C}$ a set of m linear constraints over $\mathcal{V}$ with p-box *cdf*-interval coefficients. A p-box *cdf*-interval Linear System *UILS* induced by $\mathcal{C}$ over $\mathcal{V}$ is a tuple $\langle \mathbf{A}, \mathbb{R}, \mathbf{b} \rangle$, where $\mathbf{A} \in (\mathbb{R} \times [0..1] \times \mathbb{R}^+)^{m \times n}$ is the matrix of the LHS intervals $[glb_{A_i}, lub_{A_i}]$, $\mathbf{b} \in (\mathbb{R} \times [0..1] \times \mathbb{R}^+)^m$ is the RHS vector of intervals $[glb_{b_i}, lub_{b_i}]$ and $\mathbb{R}_i \in \{\subset_{\mathcal{R}_U}, \subseteq_{\mathcal{R}_U}, =, \supset_{\mathcal{R}_U}, \supseteq_{\mathcal{R}_U}\}$ $\forall i = 1, \dots, m$ is a list of m relations.

Theorem 1. The Positive Orthant Linear transformation of the p-box *cdf*-intervals (POLI-CDF) $\rho = \langle \mathbf{A}, \mathbb{R}, \mathbf{b} \rangle$ is the complete solution set of the interval linear

inequality system $\langle \mathbf{A}'X\{\mathcal{R}\}\mathbf{b}' \rangle$ and it is defined on the *cdf*-interval points as follows:

$$(A_i', b_i') = \begin{cases} (glb_{A_i}, lub_{b_i}) & \text{if } \{\prec, \preceq\} \in \mathcal{R}_i \\ (lub_{A_i}, glb_{b_i}) & \text{if } \{\succ, \succeq\} \in \mathcal{R}_i \end{cases}$$

Solving the produced list of constraints yields the full closure of the original CSP problem along with its maximum and minimum probability of occurrence. We omitted the formal proof of models equivalence due to limited space.

Consider a typical minimization problem over p-box *cdf*-coefficients and variables. The transformation of the model yields (2^{n+1}) inequalities. The produced solution set is $S_i = \{S_i^k | k = 1, 2, \dots, 2^{n+1}\}$ where the upper-bound value range is $S_i = lub_{k=1}^{2^{n+1}} S_i^k$ and the lower-bound value range is $S_i = glb_{k=1}^{2^{n+1}} S_i^k$.

The algorithm steps 1-3 are similar to those adopted in [24] and yields the same quantile results. The last equality constraint of example 4 is equivalently rewritten as two constraints:

$$\begin{aligned} (1.5, 0.14, 0.96)X1 +_{\mathcal{R}_U} (5.999, 0.28, 0.86)X2 &\subseteq_{\mathcal{R}_U} (15.0, 0.85, 0.029) \\ \wedge (3.0, 0.96, 0.034)X1 +_{\mathcal{R}_U} (6.0, 0.86, 0.028)X2 &\supseteq_{\mathcal{R}_U} (4, 0.4, 0.088) \end{aligned}$$

Project the cdf-distributions and extract the final solution bounds The set of linear equations $\mathbf{AX} = \mathbf{b}$ can be rewritten $\mathbf{X} = \mathbf{b}/\mathbf{A}$. Domains of variables in each addition constraint will be pruned based on the result of the set of *cdf*-divisions $\mathbf{b}/\mathbf{A}$. In the worst case scenario $(2n)$ *cdf*-divisions per constraint will take place. We multiply by 2 since each bound per variable requests one operation. Points for each variable are the quantile values obtained from the 1st dimension operation along with the list of lines output from the exerted *cdf*-divisions. We project the quantile values onto the line by getting its corresponding *cdf* value and slope to compose the triplet point structure. We then seek the least upper bound and greatest lower bound for all resulting points per variable. This operation yields the variable new p-box *cdf*-interval pruned domain.

Performance Comparison The framework was tested on two systems of constraints with densities: 12 and 127 variables. Performance is given in terms of the real-time taken to generate the bounds of the final solution set. We used the ECL'PS^e profiling in order to calculate the percentage of the total time each predicate takes to carry-out its designated task. The experiment was performed on a Core2 Duo CPU, 2.53GHz and 3GB RAM under a windows environment.

Real-time performance of the p-box *cdf*-intervals framework is compared to the uncertain CSP (UCSP) [24] and the original *cdf*-intervals model. Coefficient bounds of the ILS are chosen to be the same to check for the ability of the algebraic structure to shrink the bounds of the solution set. Obviously, as shown in Table 1 UCSP is less expensive because the realization of the solution is exerted on the real-bound points only. Information within the interval bounds has equally weighted probability distribution. In that case no propagation on the probability bounds takes place. On the other hand the original *cdf*-intervals

	12 variables			127 variables		
	p-box <i>cdf</i>	UCSP	<i>cdf</i> -intervals	p-box <i>cdf</i>	UCSP	<i>cdf</i> -intervals
Total Real Time	0.405	0.358	0.577	94.64	86.45	195.09
Generate extended linear inequalities	7.69%	8.69%	5.4%	21.29%	25.26%	10.13%
Extract quantile ILS	3.84%	4.34%	5.40%	11.42%	12.39%	10.48%
Solve 2n eplex instances	80.76%	86.95%	89.18%	57.14%	62.34%	79.37%
Project the <i>cdf</i> -distributions	7.69%	N/A	N/A	10.13%	N/A	N/A

Table 1. Real-time execution

framework has double the number of constraints and eplex instances produced. This algebraic structure seeks the propagation of the bounds in the two dimensions: quantile and *cdf*. Clearly most of the time spent in all frameworks is the time taken to solve the $2n$ eplex instances. This time for the p-box *cdf*-intervals and UCSP is about 80% and 60% for the small and medium density problem instances respectively. On the other hand it takes about 90% and 80% of the real-time spend by the original *cdf*-intervals approach. Typically in the p-box *cdf*-intervals division and projection are linear operations. Such operations take about 10% of the time spent to propagate the solution extreme points.

5 Case Study: Network Traffic Analysis Problem (NTAP)

In telecommunication the Traffic Matrix TM is typically used in network design problems to seek the realization of all possible flows by installing links at minimum cost while satisfying the link capacity constraints (i.e. the flow cannot exceed the link installed capacity). Exploiting TM with complete OD pair combinations for a real-world large network is an expensive computation. This computation is directly proportional to the square of the matrix dimension: the number of OD pair entries. TM data is obtained through link-counts which is measured by the Simple Network Management Protocol (SNMP) [5, 15].

However, traffic volumes in networks fluctuate; those fluctuations push network planners to overestimate the flow in order to ensure adequate network sustainability. Statistical methods seek to estimate demands based on prior measurements. Robust techniques formulate the network problem in mixed-integer programming and compute minimum and maximum bounds that encapsulate all feasible realizations of the end-to-end network traffic flow. Estimated resulting flow has an equal weight of uncertainty and lack the portrayal of data whereabouts [16].

Constructing p-box for uncertain data To use our framework we need to model the data uncertainty using a p-box *cdf*-interval. Given an observed data set, an upper and lower uniform *cdf* bounds that enclose all observed data values are computed based on the quantile values and the set of respective *cdfs*. The algorithm is a preprocessing step and does not cause overhead. It is performed once for each uncertain data. The algorithm has $O(n \log(n))$ complexity where n is the number of distinct data measured. To obtain the slope of the lower bound point, we draw the steepest upper bound slope from the first point and the flattest

lower bound slope from the last point. This action guarantees the envelopment of all measured data between two lines: in this case upper and lower *cdf* uniform distribution bounds.

Computing p-box data intervals for the NTAP To generate and convert the measured data into *cdf*-intervals we used Matlab as an adequate mathematical tool that provides built-in function to generate and deal with different probability distributions. Raw data is provided in the form of either a single demand matrix or a set of n distinct measured-demand matrixes. In the single matrix case the demand is approximated, whereas in the case of multiple demand matrixes each matrix represents a demand measurement at a given time point. [17, 18] provides both types of datasets: single and dynamic set of traffic matrixes.

Comparison between the data models We compare three uncertain data models: 1. the p-box *cdf*-Intervals 2. the original *cdf*-Intervals 3. UCSP, with bounds formed using an error correction model ($\mu, 40$)

We have generated 80 instances of a 4-nodes network problem: 40 instances based on Poisson distributions and the other 40 are based on Normal distributions of OD-pairs. Link-counts were simulated. Output solution sets of all maneuvers witnessed a total encapsulation of the generated Netflow data. The p-box *cdf*-intervals can further prune the output solution domains to maintain interval consistency properties in the two dimensions i.e. prohibit the upper bound probability from being dominated by lower bound distribution. The solver detected this scenario in 33% -42% of the cases when Poisson distribution was employed; and 16% - 25% of the cases were observed when Normal distribution was taken under consideration.

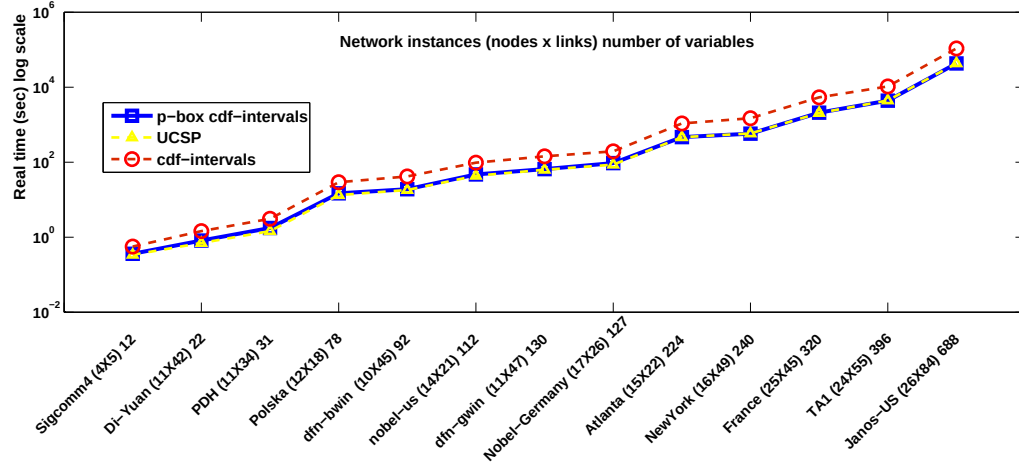


Fig. 7. Real Time Comparison

Scalability Test In this experiment we employed the corpus available in [17, 18]. As shown in Fig. 7 twelve networks with varying densities were employed: 1. Number of variables: (12 - 688) 2. Number of nodes: (4 - 26) 3. Number of links: (5 - 84) We observed that the real time to find the problem solution domains is directly proportional to the network density. The problem density is specified by the number of unknown variables the solver needs to exploit based on the given link-counts.

Fig. 7 illustrates the real time spent by the three solvers in seconds. Recall from section 4.2 the time spent by the solver is mostly utilized in solving the generated $2n$ eplex instances. The number of eplex instances in the original *cdf*-intervals is doubled; in the *cdf*-intervals algebraic structure the same number of constraint equations is produced in two dimensions; constraint equations exploit variable domain bounds per dimension; this fact explains the extra real-time the algebraic structure spend compared to the other two solvers. Clearly, Fig. 7 shows that p-box *cdf*-intervals take almost the same time as UCSP to exploit variable solution domains. Hence, our novel framework is cost effective and does not add up computation cost for large problems. Moreover, it adds up information about the minimum and maximum distribution of data; accordingly, user (domain expert) can choose to further constraint the output domain of the variable by specifying a probability limit to restrict the search space.

6 Conclusion

Convex models are favored for their tractable computation. They use interval coefficients to characterize ill-defined data, which adopt interval reasoning to seek output solution sets that satisfy all possible realization of data sought. Drawback of convex structures reveal in their solution set property which has an equal weight of knowledge. To overcome this problem, we quantified knowledge by means of *cdf*-intervals in a previous work. Due to its natural monotonic property *cdf* forms a lattice thus aggregated information is stored in the second dimension of the interval bounds. The employed *cdf* in the model approximated the probability to the nearest uniform distribution. In this paper we extended *cdf*-intervals by bounding the unknown probability distribution to guarantee the full encapsulation of data along with its degree of knowledge. We demonstrated the novel algebraic structure and the calculus behind. We supported our computation by a case study using the consortium of NTAP problems. Experimental results show that the novel framework encapsulate the behavior of the data whereabouts. Additional information conveyed allows the domain expert to further constraint the output domain of the variable by specifying a probability limit to restrict the search space. The novel model employs such behavior in a cost effective manner without adding significant computation cost for large problems.

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